الطالب الأكاديمي
THE ACADEMIC STUDENT

Faculty of Business Studies

BUS101
Introduction to Math for Business

Final Examination / Version: B
2025-2026 / Fall

17 – January – 2026

Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed: 2 hours**

tructions:

Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
Mobile phones must be in "SWITCHED **OFF"** mode throughout the examination.
This is a **CLOSED** book examination. NO documents of any kind are allowed.
You **MUST** answer the questions using only the answer booklets provided
You **MUST** return the examination paper together with your answer booklets.
Do **NOT** write your answers in pencil except graphs/sketches
Students are **NOT** allowed to use any type of dictionary or mobile phones during the exan
Students are allowed to use ***non-programmable*** calculators only (if needed).
Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy

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Question 1: Choose the correct answer (10 marks)

1. A bakery buys cupcakes for $1.20 each and sells them for $2.00. What is the amount of markup?

A. $0.70
B. $0.80
C. $0.90
D. $1.00

2. A retailer uses a 60% markup on cost. What is the equivalent markup on selling price?

A. 60%
B. 45%
C. 37.5%
D. 50%

3. A treadmill priced at $1,500 is marked down by 20%. What is the reduced price?

A. $1,250
B. $1,100
C. $1,200
D. $1,300

4. You invest $10,000 for 1 year at 6% interest. How much interest will you earn?

A. $600
B. $550
C. $580
D. $620

5. You borrow $4,000 for 5 months at 12% simple interest. Find the interest.

A. $240
B. $200
C. $180
D. $160

6. A student deposits $5,000 for 90 days and earns $62.50 interest. What is the annual interest rate? (360-da year)

A. 5%
B. 4.5%
C. 6%
D. 4%

7. A deposit of $15,000 earns $300 in interest at 6% annually. How many days did the money earn interest 360-day year)

A. 120 days
B. 90 days
C. 100 days
D. 60 days

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8. For 8% interest compounded semiannually for 3 years, what is the rate per period and number of periods?

A. 4%, 6 periods
B. 8%, 3 periods
C. 2%, 4 periods
D. 4%, 3 periods

9. Find the rate and periods for 5% compounded quarterly for 4 years.

A. 1.25%, 16 periods
B. 5%, 4 periods
C. 2%, 10 periods
D. 1%, 20 periods

10. A customer makes 24 payments of $80 and pays $200 down for a washing machine. What is the total installment cost?

A. $2,100
B. $2,000
C. $2,150
D. $2,120

Question 2: (8 marks)

A. A smartphone is sold for $1,200 and the markup is $240. What is the markup percentage based on selling price? (4 marks)

B. A chair is marked up by $180 and has a markup of 60% on cost. Find the cost and selling price. (4 marks)

Question 3: (8 marks)

A. A game console sells for $499 and cost the store $375. Calculate the markup and markup percent base on selling price. (4 marks)

B. A bakery bakes 150 loaves of bread for $2.20 each. 20% are wasted. If they want 25% markup on selling price, find the price per loaf. (4 marks)

Question 4: (8 marks)

A. A speaker system costs $350 and expenses are 12% of the cost. If sold for $400, calculate Breakeven (BE) point and profit\loss amount.

(4 marks)

B. A tablet costs $300 and is sold at a markdown of 20% from $450. Operating expenses are 25% of co Calculate the breakeven (BE) point, operating loss, and absolute loss.

(4 marks)

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Question 5: (8 marks)

A. A logistics company borrows $55,000 at 6.2% annual interest on February 1 for 200 days. A payment of $20,000 is made on April 15. Find the balance owed on the principal after the payment. (4 marks)

B. Find the interest earned on a $1,800 deposit for 3 years, compounded quarterly at 6%. (4 marks)

Question 6: (8 marks)

a) A school board plans to renovate its science labs in 6 years at a cost of $2,400,000. They decide to make end-of-quarter deposits into a sinking fund earning 6% interest compounded quarterly. Find: (4 marks)

(a) the amount of each quarterly payment
(b) the interest earned

b) Lisa purchased a home gym system for $1,750. She paid $250 down and financed the rest with 24 monthly payments of $65.25. (4 marks)
Find:
(a) the total installment cost,
(b) the finance charge,
(c) the amount financed.

End of the questions

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1½%

PERIOD	TABLE 12-1 COMPOUND VALUE OF $1	TABLE 12-3 PRESENT VALUE OF $1	TABLE 13-1 AMOUNT OF ANNUITY OF $1	TABLE 13-2 PRESENT VALUE OF ANNUITY OF $1	TABLE 13-3 SINKING FUND VALUE OF $1
1	1.0150	0.9852	1.0000	0.9852	1.0000
2	1.0302	0.9707	2.0150	1.9559	0.4963
3	1.0457	0.9563	3.0452	2.9122	0.3284
4	1.0614	0.9422	4.0909	3.8544	0.2444
5	1.0773	0.9283	5.1522	4.7826	0.1941
6	1.0934	0.9145	6.2295	5.6972	0.1605
7	1.1098	0.9010	7.3230	6.5982	0.1366
8	1.1265	0.8877	8.4328	7.4859	0.1186
9	1.1434	0.8746	9.5593	8.3605	0.1046
10	1.1605	0.8617	10.7027	9.2222	0.0934
11	1.1780	0.8489	11.8632	10.0711	0.0843
12	1.1960	0.8364	13.0412	10.9075	0.0767
13	1.2135	0.8240	14.2368	11.7315	0.0702
14	1.2318	0.8119	15.4503	12.5433	0.0647
15	1.2502	0.7999	16.6821	13.3432	0.0599
16	1.2690	0.7880	17.9323	14.1312	0.0558
17	1.2880	0.7764	19.2013	14.9076	0.0521
18	1.3073	0.7649	20.4893	15.6725	0.0488
19	1.3270	0.7536	21.7966	16.4261	0.0459
20	1.3469	0.7425	23.1236	17.1686	0.0432
21	1.3671	0.7315	24.4704	17.9001	0.0409
22	1.3876	0.7207	25.8375	18.6208	0.0387
23	1.4084	0.7100	27.2250	19.3308	0.0367
24	1.4295	0.6995	28.6334	20.0304	0.0349
25	1.4510	0.6892	30.0629	20.7196	0.0333
26	1.4727	0.6790	31.5138	21.3986	0.0317
27	1.4948	0.6690	32.9866	22.0676	0.0303
28	1.5172	0.6591	34.4813	22.7267	0.0290
29	1.5400	0.6494	35.9986	23.3760	0.0278
30	1.5631	0.6398	37.5385	24.0158	0.0266
31	1.5865	0.6303	39.1016	24.6461	0.0256
32	1.6103	0.6210	40.6881	25.2671	0.0246
33	1.6345	0.6118	42.2984	25.8789	0.0236
34	1.6590	0.6028	43.9329	26.4817	0.0228
35	1.6839	0.5939	45.5919	27.0755	0.0219
36	1.7091	0.5851	47.2758	27.6606	0.0212
37	1.7348	0.5764	48.9849	28.2371	0.0204
38	1.7608	0.5679	50.7197	28.8050	0.0197
39	1.7872	0.5595	52.4805	29.3645	0.0191
40	1.8140	0.5513	54.2677	29.9158	0.0184
41	1.8412	0.5431	56.0817	30.4589	0.0178
42	1.8688	0.5351	57.9229	30.9940	0.0173
43	1.8969	0.5272	59.7917	31.5212	0.0167
44	1.9253	0.5194	61.6886	32.0405	0.0162
45	1.9542	0.5117	63.6139	32.5523	0.0157
46	1.9835	0.5042	65.5681	33.0564	0.0153
47	2.0133	0.4967	67.5516	33.5531	0.0148
48	2.0435	0.4894	69.5649	34.0425	0.0144
49	2.0741	0.4821	71.6084	34.5246	0.0140
50	2.1052	0.4750	73.6825	34.9996	0.0136

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3%

PERIOD	TABLE 12-1 COMPOUND VALUE OF $1	TABLE 12-2 PRESENT VALUE OF $1	TABLE 13-1 AMOUNT OF ANNUITY OF $1	TABLE 13-2 PRESENT VALUE OF ANNUITY OF $1	TABLE 13-3 SINKING FUND VALUE OF $1
1	1.0300	0.9709	1.0000	0.9709	1.0000
2	1.0609	0.9426	2.0300	1.9135	0.4926
3	1.0927	0.9151	3.0909	2.8286	0.3235
4	1.1255	0.8885	4.1836	3.7171	0.2390
5	1.1593	0.8626	5.3091	4.5797	0.1884
6	1.1941	0.8375	6.4684	5.4172	0.1546
7	1.2299	0.8131	7.6625	6.2303	0.1305
8	1.2668	0.7894	8.8923	7.0197	0.1125
9	1.3048	0.7664	10.1591	7.7861	0.0984
10	1.3439	0.7441	11.4639	8.5302	0.0872
11	1.3842	0.7224	12.8078	9.2526	0.0781
12	1.4258	0.7014	14.1920	9.9540	0.0705
13	1.4685	0.6810	15.6178	10.6350	0.0640
14	1.5126	0.6611	17.0863	11.2961	0.0585
15	1.5580	0.6419	18.5989	11.9379	0.0538
16	1.6047	0.6232	20.1569	12.5611	0.0496
17	1.6528	0.6050	21.7616	13.1661	0.0460
18	1.7024	0.5874	23.4144	13.7535	0.0427
19	1.7535	0.5703	25.1169	14.3238	0.0398
20	1.8061	0.5537	26.8704	14.8775	0.0372
21	1.8603	0.5375	28.6765	15.4150	0.0349
22	1.9161	0.5219	30.5368	15.9369	0.0327
23	1.9736	0.5067	32.4529	16.4436	0.0308
24	2.0328	0.4919	34.4265	16.9355	0.0290
25	2.0938	0.4776	36.4593	17.4131	0.0274
26	2.1566	0.4637	38.5530	17.8768	0.0259
27	2.2213	0.4502	40.7096	18.3270	0.0246
28	2.2879	0.4371	42.9309	18.7641	0.0233
29	2.3566	0.4243	45.2188	19.1885	0.0221
30	2.4273	0.4120	47.5754	19.6004	0.0210
31	2.5001	0.4000	50.0027	20.0004	0.0200
32	2.5751	0.3883	52.5027	20.3888	0.0190
33	2.6523	0.3770	55.0778	20.7658	0.0182
34	2.7319	0.3660	57.7302	21.1318	0.0173
35	2.8139	0.3554	60.4621	21.4872	0.0165
36	2.8983	0.3450	63.2759	21.8323	0.0158
37	2.9852	0.3350	66.1742	22.1672	0.0151
38	3.0748	0.3252	69.1594	22.4925	0.0145
39	3.1670	0.3158	72.2342	22.8082	0.0138
40	3.2620	0.3066	75.4012	23.1148	0.0133
41	3.3599	0.2976	78.6633	23.4124	0.0127
42	3.4607	0.2890	82.0232	23.7014	0.0122
43	3.5645	0.2805	85.4839	23.9819	0.0117
44	3.6715	0.2724	89.0484	24.2543	0.0112
45	3.7816	0.2644	92.7198	24.5187	0.0108
46	3.8950	0.2567	96.5014	24.7754	0.0104
47	4.0119	0.2493	100.3965	25.0247	0.0100
48	4.1323	0.2420	104.4084	25.2667	0.0096
49	4.2562	0.2350	108.5406	25.5017	0.0092
50	4.3839	0.2281	112.7968	25.7298	0.0089

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